

Basic Movement — What each column means

1) Movement (per-frame displacement)

What it measures: how far each body part moves between consecutive frames.

- **{bp}_movement**
Distance that body part **bp** traveled since the previous frame.
Units: mm/frame (we convert from pixels using your px_per_mm).
- **Summed_movement**
Frame-by-frame **total movement** across **all** tracked body parts (a simple “activity” signal).
Units: mm/frame.
- **{bp}_movement_{W}_mean**
Average movement of **bp** over the last **W** frames.
Units: mm/frame.
- **{bp}_movement_{W}_sum**
Total movement of **bp** over the last **W** frames.
Units: mm.

Body parts used: all parts listed in your project’s bp_names (e.g., bodypart1 ... bodypart16).

You get one {bp}_movement column per body part, plus the single Summed_movement.

2) Velocity (speed proxy)

What it measures: how much distance a body part covered in the **last second**.

- **{bp}_velocity**
Rolling sum of {bp}_movement over **1 second** (i.e., fps frames). Numerically, that’s the distance covered in the last second.
Units: mm in the last 1 s (commonly treated as **mm/s**).
- **Bp_velocity_mean**
The **average** of all {bp}_velocity values at a frame (one whole-fish speed trace).
Units: mm in the last 1 s (≈ mm/s).
- **Bp_velocity_stdev**
The **spread** (standard deviation) of {bp}_velocity across body parts.
Units: same as above.

- **Minimum_avg_bp_velocity_{W}_window**
Over the past **W** frames, the **minimum** value of Bp_velocity_mean.
Units: same as Bp_velocity_mean.
- **Max_avg_bp_velocity_{W}_window**
Over the past **W** frames, the **maximum** value of Bp_velocity_mean.
Units: same as Bp_velocity_mean.
- **Absolute_diff_min_max_avg_bp_velocity_{W}_window**
The range: Max – Min of Bp_velocity_mean over the last **W** frames.
Units: same as Bp_velocity_mean.

Body parts used: all parts (one {bp}_velocity per part, then aggregated).

*If you prefer a single landmark (e.g., “velocity was computed from **bodypart2**”), just read from bodypart2_velocity instead of Bp_velocity_mean.*

3) Acceleration (change in speed)

What it measures: how much a body part’s 1-second velocity has changed over your chosen window.

- **{bp}_acceleration_{i}_window**
Change in {bp}_velocity compared to **i** frames earlier:
current_velocity – velocity_shifted_by_i.
Units: mm in the last 1 s (a difference).
- **Bp_acceleration_mean_{i}_window**
Mean of {bp}_acceleration_{i}_window across **all** body parts.
Units: mm in the last 1 s.
- **Bp_acceleration_stddev_{i}_window**
Standard deviation across body parts.
Units: mm in the last 1 s.
- **Min_avg_bp_acceleration_{i}_window, Max_avg_bp_acceleration_{i}_window, Absolute_diff_min_max_avg_bp_acceleration_{i}_window**
Rolling summaries of Bp_acceleration_mean_{i}_window over the last **i** frames (min, max, and range).
Units: mm in the last 1 s.

Body parts used: all parts (per-part deltas, then aggregated).

*Again, if you want acceleration from a single landmark (e.g., **bodypart2**), use its {bp}_acceleration_{i}_window column directly and (optionally) convert to mm/s² by dividing by i/fps.*